Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/09/2006
PI Name:	Outlaw, William H Ph.D.		
Project Title:	Perturbation of the regulatory apoplastic photosynthate pool in leaves		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	32306-4370	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	03/01/2004	**End Date:**	02/28/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNA04CC68G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Prolonged spaceflight may require plant life support systems that can use light to provide food, oxygen, and recycled water for other organisms onboard. I proposed to study an organism level physiological change caused by the space environment to which plants may respond through intercellular signaling. Specifically, the space environment is CO2 enriched with respect to Earth, a condition that is expected to perturb the size and distribution of the regulatory foliar apoplastic photosynthate pool in apoplastic phloem loaders (as far as is known, all the plants on the ALS Crop List). The proposed work investigates a hypothesized link between the photosynthate concentration in this pool and the accumulation of photosynthate in the guard cell wall (which is known to diminish stomatal conductance). In a parallel comparative biological approach, a symplastic phloem loader, which is hypothesized to lack a regulatory apoplastic foliar photosynthate pool, will be studied, as these plants may be better suited for space farming. From these data, extrapolations of performance by plants during prolonged spaceflights will be made. Therefore, the results of the		

	proposed research will provide a foundation to the Fundamental Space Biology Division's aim of developing basic knowledge down to the cellular level that would permit long term human presence in space.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed research is basic (using state of the art methods for quantification of subcellular pools of photosynthate in a physiological context) and develops a novel field of inquiry (the role of the apoplast in integrating leaf functions in symplastic phloem loaders) and therefore will provide a better understanding of plant function on Earth. The results also will be relevant to predictions regarding climate change scenarios.
Task Progress:	Overall, we seek to learn how regulatory apoplastic solutes influence stomatal aperture size. Part of our effort focuses on the osmotic effect of apoplastic photosynthate. In apoplastic phloem loaders, photosynthate is released from parenchyma into the intercellular aqueous space from which it is accumulated by the phloem. We observed that an osmotically significant concentration of sucrose accumulates in the guard-cell apoplast during the photoperiod and subsequently proved by compartmental analysis that this sucrose is a product of recent photosynthesis in the mesophyll. We hypothesized that sucrose accumulation in the guard-cell apoplast is a function of transpiration rate and the concentration of photosynthate in the leaf apoplast (which itself is the balance of photosynthate release by parenchyma and its translocation from the leaf). The first part of this hypothesis was proven by a diminution of sucrose accumulation in the guard cell apoplast when the driving force for transpiration was reduced four-fold without a compensatory increase in stomatal conductance. At the beginning of this award, we had not addressed the second part of the hypothesis, viz., photosynthate concentration in the apoplast. Also, symplastic phloem loaders, which obviously serve as a control for apoplastic loaders in the context of our hypothesis, had not been used for stomatal studies. The award was for one year (Mar 01 2004 – Feb 28 2005) and we received a no-cost extension to finesse some aspect of the work and begin preparation of manuscripts. The one-year award called for (a) determination whether the sucrose concentration of the leaf apoplast is a function of CO2 concentration in the short term for apoplastic phloem loaders; (b) identification of a symplastic model species for stomatal studies followed by its (i) physiological characterization (diel kinetics of stomatal aperture size, guard-cell starch content, guard-cell K+ content, and photosynthate concentration in the leaf apoplast) and (ii) demonstration that guard cells can be microdissected; (c) and attempt to identify the apoplastic pool sampled by the pressure chamber and prove that symplastic contents are not released during pressurization. All of these goals, and more, were accomplished with the exception of a subset of experiments for (c) above. Two manuscripts are in the final stages of preparation. These manuscripts report results of the proposed work and much more. In the detailed progress report, find the results that will be submitted for publication. In the first manuscript, we lowered the photosynthetic rate of an apoplastic phloem loader by diminishing light and CO2 concentration without altering conductance. As expected, the mobile pool of photosynthate in the bulk apoplast was decreased. Then, we determined that under these conditions, the stomatal aperture size was not diminished osmotically by transpiration-linked solute accumulation in the guard-cell apoplast. This key result was the final proof needed to support the theory of partial control of conductance in apoplastic phloem loaders by the integrated effects of photosynthesis and water loss (see para 1) In the second manuscript, we report characterization of a symplastic loading species that is amenable to stomatal work and quantitative histochemistry. Then, we used this species to show that partial control of conductance in symplastic phloem loaders by the integrated effects of photosynthesis and water loss does not occur.
Bibliography Type:	Description: (Last Updated:)